Sparsh Sunil Naik

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SUMMARY

I build real-world AI systems focused on decision-making and perception under uncertainty. My work includes reinforcement learning for dynamic pricing and vision-based control systems, with a focus on translating models into deployable, reliable pipelines.

EDUCATION

KLE Technological University — **2023 – Present**
B.E. in Computer Science and Engineering (Artificial Intelligence) — *CGPA: 8.99 (after 4th semester)*

Arjuna Science PU College, Dharwad, Karnataka — **2021 – 2023**
Science Stream — *PCM: 92%, KCET: 8492*

PROJECTS

Vision-Language-Action (VLA) System for Robotics — **Present**

* Developing a real-time system that maps visual input to action decisions using vision-language models and control logic.
* Designed perception-to-action pipelines integrating camera input, feature extraction, and decision-making modules.
* Focused on latency, robustness, and real-world deployment constraints in dynamic environments.

Dynamic Pricing using Reinforcement Learning — **09/2025 - 12/2025**

* Designed a reinforcement learning environment to model retail pricing under uncertain demand, inventory constraints, and competing sellers.
* Implemented and compared SAC, PPO, DDPG, and contextual bandit baselines using a shared simulation framework.
* Evaluated policies using profit, price volatility, and stockout rate to study stability–performance tradeoffs.
* (Paper accepted, publication expected soon)

Smart Helmet Accident Detection using IoT and Edge Analytics — **09/2025 - 12/2025**

* Designed an IoT-based safety system using multi-sensor fusion (accelerometer, gyroscope, GPS) for accident detection.
* Built an edge-processing pipeline to reduce latency and false positives before cloud transmission.
* Evaluated system performance under sensor noise and threshold variations.

WiDS Datathon – Machine Learning Modeling Pipeline — **02/2025 - 05/2025**

* Developed an end-to-end machine learning pipeline for a real-world prediction task as part of the WiDS Datathon.
* Implemented and evaluated models including XGBoost, LightGBM, and Logistic Regression with validation-based selection.
* Achieved an F1 score of 0.78 on the official leaderboard, securing a global rank of 17.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, JavaScript
Machine Learning & Computer Vision: NumPy, Pandas, Matplotlib, Scikit-learn; reinforcement learning (SAC, PPO), model evaluation (F1, cross-validation, hyperparameter tuning)
Data Processing: Feature preprocessing, normalization, handling class imbalance
Systems & Networking: IoT pipelines, MQTT, edge computing concepts, sensor data processing
Web & Backend: Flask, HTML, CSS, JavaScript
Databases: SQL, MongoDB
Development Environments: Linux (Ubuntu-based), Vim, Git, GitHub, VS Code, Google Antigravity
Relevant Coursework: Data Structures and Algorithms, Discrete Mathematics, Probability, Linear Algebra, Computer Networks, Operating Systems

POSITIONS OF RESPONSIBILITY

Data Science Club - President **2025 – Present**

- Led EDA workshops and organized datathons using custom datasets to encourage applied problem-solving.

Cloud and DevOps Club - Core Team Member **2025 – Present**

- Contributed to technical sessions on cloud infrastructure, DevOps workflows, and systems fundamentals.